B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

IT 19-204-0403 OPERATING SYSTEMS
(2019 Scheme)

Time: 3 Hours — Maximum Marks: 60

Course Outcomes
On successful completion of the course, the students will be able to:

CO1: Design various scheduling algorithms.
CO2: Apply the principles of concurrency.
CO3: Design deadlock, prevention and avoidance algorithms.
CO4: Compare and contrast various memory management schemes.
CO5: Design and implement a prototype file system.
CO6: Attain basic knowledge about Real time operating systems.
Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

PART A
(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	What is the primary purpose of an operating system and how does it provide an environment for executing programs?	3	L2	1	1,2
	(b)	Define race conditions and explain why they are problematic in concurrent systems.	3	L1, L2	2	1,2
	(c)	How do bitmaps track memory usage and what challenges does it pose in terms of scalability and fragmentation?	3	L2	4	1,2
	(d)	Explore the challenges and trade-offs involved in designing a segmentation system.	3	L4	4	2,3
	(e)	Characterize a system that is in a deadlock state. What are the observable symptoms and how can we identify such a state?	3	L4	3	2,3
	(f)	Explore strategies for deadlock recovery.	3	L2	3	2,3
	(g)	How do access control lists (ACLs) and file permissions contribute to security?	3	L2	5	2,3
	(h)	What distinguishes an RTOS from a general-purpose operating system?	3	L2	6	2,3

PART B

(4 × 12 = 48)

			Marks	BL	CO	PO
II.	(a)	Describe how semaphores are used to implement process synchronization. What are the wait (P) and signal (V) operations?	4	L2	2	1,2
	(b)	Given below are the arrival and burst times of four processes P1, P2, P3 and P4. Draw the Gantt chart using FCFS, SJF (pre-emptive), SJF (non-pre-emptive) and RR scheduling (Quantum = 4 ms, no priority based pre-emption). Calculate the average waiting time, average turnaround time and average response time.	8	L3	1	

Process	Arrival time (msecs)	Burst time (msecs)
P1	0	8
P2	1	4
P3	2	9
P4	3	5

OR

			Marks	BL	CO	PO
III.	(a)	Explain the concept of system calls and their role in facilitating interactions between user programs and the operating system.	6	L2	1	1,2,3
	(b)	What is a critical section and why is mutual exclusion necessary to prevent multiple processes from accessing it simultaneously?	6	L2	2	
IV.	(a)	Explain the concept of swapping in memory management. How does it enhance multiprogramming?	4	L2	4	1,2,3
	(b)	Consider the following reference string 0 1 5 3 0 1 4 0 1 5 3 4 and the number of frames in the memory to be 3. Find the number of page faults respective to (i) Optimal Page Replacement Algorithm (ii) FIFO Page Replacement Algorithm (iii) LRU Page Replacement Algorithm.	8	L3	4	
		OR				
V.	(a)	Describe the Buddy system for memory allocation. How does it handle requests for memory blocks of varying sizes and what are its limitations?	6	L2	4	1,2,3
	(b)	Explain Belady's Anomaly with the example of the following reference string: 0 1 5 3 0 1 4 0 1 5 3 4.	6	L3	4	
VI.		Consider a system with three resource types (A, B and C) and five processes (P1, P2, P3, P4 and P5). The maximum resource requirements for each process are as follows. P1: A(7), B(5), C(3) P2: A(3), B(2), C(2) P3: A(9), B(0), C(2) P4: A(2), B(2), C(2) P5: A(4), B(3), C(3) The current allocation of resources is: P1: A(1), B(0), C(0) P2: A(0), B(1), C(0) P3: A(3), B(0), C(2) P4: A(2), B(1), C(1) P5: A(0), B(0), C(2) The available resources are: A(3), B(3), C(2) Use the Banker's algorithm to determine whether granting the following resource requests will lead to a safe state: (i) Process P1 requests A(2), B(1), C(0). (ii) Process P3 requests A(0), B(0), C(2).	12	L3	3	1,2,3
		OR				
VII.	(a)	Discuss the concept of resource trajectories in deadlock avoidance. How can we predict whether granting a resource request will lead to a safe or unsafe state?	6	L2	3	2,3
	(b)	Discuss the concept of starvation in resource allocation. How can we prevent certain processes from being perpetually denied access to resources?	6	L2	3	

(Continued)

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			Marks	BL	CO	PO
VIII.	(a)	Explain the difference between contiguous allocation, linked allocation and indexed allocation in file systems.	4	L2	5	1,2,3
	(b)	Consider a disk with 200 tracks and the queue has random requests from different processes in the order: 55, 58, 39, 18, 90, 160, 150, 38, 184 Initially arm is at 100. Find the Average Seek length using FIFO, SSTF, SCAN and C-SCAN algorithm.	8	L3	6	
		OR				
IX.	(a)	What is Direct Memory Access (DMA)? How does it improve data transfer efficiency between I/O devices and memory?	6	L1	5	2,3,4
	(b)	Describe the different levels of RAID. How does RAID improve data reliability and performance?	6	L2, L5	3	

Bloom's Taxonomy Levels
L1 – 8%, L2 – 52%, L3 – 33%, L4 – 5%, L5 – 2%.

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